

Palestine Polytechnic University

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Microprocessors Project

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8085 System Memory Map

Total Address Space

8085 has 16-bit address bus → 64 KB memory space (0000H – FFFFH)

We will divide it as:

- ROM (EPROM): 8 KB total (2 × 4 KB).
- RAM: 16 KB total (2 × 8 KB).
- Memory-mapped I/O: 1 location at F000H.
- Peripheral-Mapped Port 80H.
- Remaining space: unused.

Memory Map

Address Range for the EPROM1

4K which means 12 Address lines.

0000H-0FFFH

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

0000 0000 0000 0000

0000 1111 1111 1111

Address Range for the EPROM2

4K which means 12 Address lines.

1000H-1FFFH

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

0001 0000 0000 0000

0001 1111 1111 1111

EPROM is placed at the lowest addresses because 8085 starts execution from 0000H after reset, each EPROM is 4 KB needs A0 – A11 address lines (Knowing that EPROMs are selected when A15–A13 = 000 A12 is used to distinguish between EPROM1 (0000H–0FFFH) and EPROM2 (1000H – 1FFFH)).

Address Range for RAM1

8K which means 13 Address lines.

2000H-3FFFH

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

0010 0000 0000 0000

0011 1111 1111 1111

RAM1 is enabled when the input of the decoder (3 to 8) which is A15 A14 A13
= 001

Address Range for RAM2

8K which means 13 Address lines.

4000H-5FFF

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

0100 0000 0000 0000 0000

0101 1111 1111 1111 1111

RAM2 is enabled when the input of the decoder (3 to 8) which is A15 A14 A13
= 010

Unused

Range

6000H-EFFFH

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

0110 0000 0000 0000

1110 1111 1111 1111

Left unused to simplify decoding, this ensures no memory foldback.

Memory-mapped Seven-Segment LED

Address F000H

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

1111 0000 0000 0000

D7 D6 D5 D4 D3 D2 D1 D0

0000 0000

The seven-segment display is memory-mapped at address F000H, when the CPU writes data to this address, the output latch captures the value and drives the seven-segment display.

Unused

Range

F001H-FFFFH

A15 A14 A13 A12 A11 A10 A9 A8 A7 A6 A5 A4 A3 A2 A1 A0

1111 0000 0000 0001

1111 1111 1111 1111

Peripheral-Mapped input port (DIP switches).

IN 80

Address

A7 A6 A5 A4 A3 A2 A1 A0

1000 0000

The DIP switches are connected as a peripheral-mapped input device at port 80H, the 8085 uses the IN instruction to read the port via the I/O address space, controlled by IO/M signal.